

55. Consider the inequations

$5x - 4y + 12 < 0$, $x + y < 2$, $x < 0$ and $y > 0$. Which one of the following points lies in the common region?

- (a) (0, 0)
- (b) (-2, 4)
- (c) (-1, 4)
- (d) (-1, 2)

56. Consider the following statements in respect of the function $y = [x]$, $x \in (-1, 1)$ where $[.]$ is the greatest integer function:

1. Its derivative is 0 at $x = 0.5$
2. It is continuous at $x = 0$

Which of the above statements is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

57. What is the degree of the differential

equation $1 + \left(\frac{dy}{dx}\right)^2 = \left(\frac{d^2y}{dx^2}\right)^{\frac{4}{3}}$?

- (a) $\frac{4}{3}$
- (b) 2
- (c) 3
- (d) 4

58. A radioactive substance decays at a rate proportional to the amount of substance present. If half of the substance decays in 100 years, then what is the decay constant (proportionality constant)?

- (a) $\frac{\ln 2}{100}$
- (b) $\frac{\ln 5}{100}$
- (c) $\frac{\ln 10}{100}$
- (d) $\frac{2 \ln 2}{100}$

59. What is the domain of the function

$f(x) = \sqrt{1 - (x - 1)^2}$?

- (a) (0, 1)
- (b) [-1, 1]
- (c) (0, 2)
- (d) [0, 2]

60. The area of the region bounded by the parabola $y^2 = 4kx$, where $k > 0$ and its latus rectum is 24 square units. What is the value of k ?

- (a) 1
- (b) 2
- (c) 3
- (d) 4

61. $\int_0^{\frac{\pi}{4}} \frac{dx}{(\sin x + \cos x)^2}$ किसके बराबर है ?

(a) $-\frac{1}{2}$

(b) $\frac{1}{2}$

(c) 1

(d) $\frac{3}{2}$

62. $\int (\sin x)^{-1/2} (\cos x)^{-3/2} dx$ किसके बराबर है ?

(a) $\sqrt{\tan x} + c$

(b) $2\sqrt{\tan x} + c$

(c) $\sqrt{\cot x} + c$

(d) $\sqrt{2 \tan x} + c$

63. यदि $I_1 = \int \frac{e^x dx}{e^x + e^{-x}}$ और $I_2 = \int \frac{dx}{e^{2x} + 1}$

हैं, तो $I_1 + I_2$ किसके बराबर है ?

(a) $\frac{x}{2} + c$

(b) $x + c$

(c) $\ln(e^x + e^{-x}) + c$

(d) $\ln(e^x - e^{-x}) + c$

64. $\int_{-2}^{-1} \frac{x}{|x|} dx$ किसके बराबर है ?

(a) -2

(b) -1

(c) 1

(d) 2

65. $\sin 4x + 2x$ के कितने चरममान है/हैं, जहां $0 < x < \frac{\pi}{2}$ है ?

(a) 1

(b) 2

(c) 4

(d) 8

66. फलन $f(x) = \frac{1}{\tan x + \cot x}$ का अधिकतम मान क्या है, जहां $0 < x < \frac{\pi}{2}$ है ?

(a) $\frac{1}{4}$

(b) $\frac{1}{2}$

(c) 1

(d) 2

67. यदि

$$4f(x) - f\left(\frac{1}{x}\right) = \left(2x + \frac{1}{x}\right)\left(2x - \frac{1}{x}\right) \text{ है,}$$

तो $f(2)$ किसके बराबर है ?

(a) 0

(b) 1

(c) 2

(d) 4